

called by clients and you call methods that run on the client. Similarly, in the client you define methods that can be called by the server and you call methods that run on the server. In this way, the server and clients communicate with each other. The Hub API is recommended.

**Persistent Connection API:** This is a low level API, i.e., the foundation hubs are built on top of it, and it exposes a few properties that supply entry points from where we send messages to the connected clients.

## Transport mechanisms

The transport mechanisms currently supported by SignalR are listed below, in order of preference.

**Web socket:** This is the most efficient mechanism and establishes true, persistent, two-way connections. It is supported by most modern Web browsers, the latest versions of IE, Google Chrome and Mozilla Firefox, and is only partially implemented on Safari and Opera. It makes efficient use of server memory and has the lowest latency.

**Server sent events (Event source):** This is supported by all browsers that support HTML5, except Internet Explorer.

**Forever frame:** This creates IFrame, which makes a request to an endpoint on the server that is not complied with. The server then continuously sends a script to the client which is immediately executed, providing a one-way real-time connection from server to client. This is mainly for IE.

**Long polling:** Long polling does not create a persistent connection, but instead polls the server with a request that stays open until the server responds, at which point the connection closes, and a new connection is requested immediately.

## The limitations of SignalR

- Though SignalR provides two-way communication protocols, it suffers from performance and scaling issues because of the need for better infrastructure (auto-scaling, multi-region, etc).
- When it comes to CPU utilisation with SignalR, there is an impact on the average load with thousands of clients getting connected to the server, and the performance becomes an issue.
- Sometimes it's possible that the client will not receive

## A comparison of real-time connectivity platforms

| Features     | PubNub                                                                                                                                               | Pusher                                                                              | SignalR                                                                                                                                                                                                                                             |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Licence      | Licence is required                                                                                                                                  | Licence is required                                                                 | SignalR is open source (Free to use)                                                                                                                                                                                                                |
| Capacity     | Can deal with 1 million messages a second, streams more than 3 million messages a second to 150 million devices a month with a very good performance | 40 billion monthly messages                                                         | 1000 queue requests can go through in less than 4 seconds. When the user count goes over 1000, there can be an impact on the Web server connection, the Windows thread and resource limits, eventually impacting the performance of the application |
| SDK support  | PubNub supports over 60 languages and development platforms in SDKs (.NET, JAVA, iOS, Android, Ruby, JavaScript, and Objective C, etc)               | SDK can be accessed by .NET, JAVA, iOS, Android, Ruby, JavaScript, Objective C, etc | SDK can be accessed by Android, JAVA library and C#                                                                                                                                                                                                 |
| Availability | 99.999 per cent availability guaranteed                                                                                                              | 99.999 per cent availability guaranteed                                             | Depends upon the server load and number of users                                                                                                                                                                                                    |
| Performance  | PubNub is scalable and reliable                                                                                                                      | It is less scalable in comparison to PubNub.                                        | SignalR, though providing a basic two-way communication protocol, would still have an impact on the performance and scaling issues, and also needs users to look after their own infrastructure (auto-scaling, multi-region, etc)                   |
| Cost         | PubNub's implementation cost is less in comparison to Pusher.                                                                                        | Pusher's implementation cost is more than PubNub.                                   | It is open source and accessible through GitHub.                                                                                                                                                                                                    |